

Firdaus Soberi

ORCID ID: 0000-0003-1734-0610

firdaussoberi.github.io
firdaus.soberi@cern.ch

Education

- | | |
|--|---------------------------|
| The University of Edinburgh, UK | 2022–2026 (Expected) |
| <i>PhD in Particle Physics</i> | <i>ATLAS Experiment</i> |
| <ul style="list-style-type: none">• Thesis: <i>Search for CP Violation in the Vector-Boson-Fusion Production of the Higgs Boson in the $H \rightarrow \gamma\gamma$ Decay Channel with the ATLAS Detector</i>• Supervisors: <i>Prof. Philip Clark, Prof. Victoria Martin</i> | |
| Osaka University, Japan | 2016–2018 |
| <i>MSc in Physics</i> | <i>CANDLES Experiment</i> |
| <ul style="list-style-type: none">• Thesis: <i>Measurement of β and γ Coincidence Spectrum of ^{24}Na for CANDLES Calibration System</i>• Supervisors: <i>Prof. Tadafumi Kishimoto, Prof. Sei Yoshida</i> | |
| University of Washington Seattle, USA | 2012–2015 |
| <i>BSc (Hons) in Physics</i> | |
| <ul style="list-style-type: none">• Undergraduate Supervisor: <i>Prof. Shih-Chieh Hsu (ATLAS)</i> | |

Research Experience

- | | |
|--|-----------------------------|
| ATLAS Experiment – Higgs Physics | 2022–2026 |
| <i>The University of Edinburgh</i> | <i>UK</i> |
| <ul style="list-style-type: none">• Primary analyst for the Run-3 $H \rightarrow \gamma\gamma$ analysis of CP violation in Vector-Boson Fusion, leading the development of the analysis framework, machine-learning classification, event categorisation, and systematic uncertainty evaluation.• Contributed to Run-3 vertex reconstruction performance studies for $H \rightarrow \gamma\gamma$ under higher pile-up conditions.• Developed and validated muon punch-through simulation in ATLAS fast simulation, including a flow-matching generative approach. | |
| Research Assistant | March 2019–May 2020 |
| <i>Universiti Teknologi Malaysia</i> | <i>Malaysia</i> |
| <ul style="list-style-type: none">• Developed a statistical nuclear cascade model and curated data for proton and neutron emission following Ordinary Muon Capture, contributing to radioactive isotope production studies at the MuSIC experiment. | |
| Graduate Researcher – CANDLES | October 2016–September 2018 |
| <i>Graduate School of Science, Osaka University</i> | <i>Japan</i> |
| <ul style="list-style-type: none">• Contributed to detector calibration and position-alignment studies for the CANDLES neutrinoless double beta decay experiment. | |
| Research Assistant | December 2015–June 2016 |
| <i>Universiti Malaya</i> | <i>Malaysia</i> |
| <ul style="list-style-type: none">• Worked with Prof. Hassan Abu Kassim in the Theoretical Physics group on flavor-changing processes under the Malaysian Frontier Science Grant. | |
| CERN Winter Research Experience for Undergraduates | January 2015–April 2015 |
| <i>University of Michigan in partnership with CERN</i> | <i>CERN</i> |
| <ul style="list-style-type: none">• Worked on ATLAS trigger and track-based missing transverse energy studies in $Z(\nu\nu)h(b\bar{b})$, and contributed to an anomalous quartic gauge coupling analysis. | |

Undergraduate Researcher

June 2014–August 2015

University of Washington

USA

- Studied BSM phenomenology and ATLAS physics using MadGraph aMC@NLO, including 2HDM, anomalous quartic gauge couplings, and Monte Carlo studies of the ATLAS insertable B-layer.

Other Experience

Full Stack Developer

June 2019–May 2022

CMA Solutions Sdn. Bhd.

Malaysia

- Designed and developed cross-platform mobile applications and full-stack software systems for warehouse and logistics operations, including backend APIs and database systems.
- Developed software solutions for logistics workflow automation and operational planning.

Publications

Journal Publications

- ATLAS Collaboration. (2026). Search for anomalies in vector-boson fusion production of the Higgs boson in $H(\rightarrow \gamma\gamma)jj$ events using 164 fb^{-1} of pp collision data collected at $\sqrt{s} = 13.6\text{ TeV}$ with the ATLAS detector. [arXiv:2603.20087](#) (Accepted for publication by JHEP).
- I. H. Hashim, H. Ejiri, F. Othman, F. Ibrahim, F. Soberi, N. N. A. M. A. Ghani, T. Shima, A. Sato and K. Ninomiya. (2020). “Nuclear Isotope Production by Ordinary Muon Capture Reaction”, Nucl. Instrum. Meth. A, 163749.

Proceedings

- F. Soberi, C. Jiang, P. Clark and T. Qiu. (2026). “Generative Muon Punch-Through with Flow-Matching in ATLAS”. *In preparation for CHEP 2026 proceedings*. [CHEP 2026 abstract](#).
- A. Bohare, F. Soberi and C. Jiang. (2026). “BitNet-based Quantized Neural Networks for Efficient Inference in High-Energy Physics”, DAE Symp. Nucl. Phys. **69**, 1229–1230.
- T. Novak *et al.*, ATLAS computing activity. (2026). “The ATLAS simulation R&D programme for Run 4”, PoS LHCP2025, 126.
- F. A. G. Corchia *et al.*, ATLAS computing activity. (2025). “AtlFast3: Fast Simulation in ATLAS for LHC Run 3 and Beyond”, EPJ Web Conf. **337**, 01355.
- F. Soberi, ATLAS Collaboration. (2025). “Higgs boson mass, width and CP measurements with the ATLAS detector”, PoS DIS2025 (2025), 059.
- F. Othman, I. H. Hashim, H. Ejiri, R. Razali, F. Ibrahim and F. Soberi. (2021). “Proton neutron emission model for Muon charge exchange reaction”, AIP Conf. Proc. **2319** no.1, 080004.
- S. Umehara *et al.*. (2020). “Search for neutrino-less double beta decay of ^{48}Ca -CANDLES-”, J. Phys. Conf. Ser. **1643** no.1, 012028.
- K. Tetsuno *et al.*, CANDLES. (2020). “Status of ^{48}Ca double beta decay search and its future prospect in CANDLES”, J. Phys. Conf. Ser. **1468** no.1, 012132.

Awards & Scholarships

- Full PhD Scholarship (2022–2026), awarded by MARA, a Malaysian government agency.
- Sumitomo Foundation Scholarship, Japan (2017–2018).
- Mary Gates Research Scholarship, University of Washington (2015).
- Malaysian Public Service Department Scholarship, Malaysia (2012–2015).

Schools

- Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP 2024), Nakhon Pathom, Thailand. (12–25 June 2024), “Muon Punch-through Simulation in ATLAS” (*Poster*)
- STFC HEP School 2023, University of Oxford. (3-15 September 2023), “Vertex Finding for $H \rightarrow \gamma\gamma$ in Run-3 Using Neural Networks” (*Poster*)

Research Presentations

Conferences

- CHEP 2026, Chulalongkorn University, Bangkok, Thailand. (25–29 May 2026), “Generative Muon Punch-Through with Flow-Matching in ATLAS” (*Talk*).
- Institute of Physics (IOP) Joint APP and HEPP Annual Conference 2025, University of Cambridge. (7–9 April 2025), “Test of CP-invariance of the Higgs boson in vector-boson fusion production in the $H \rightarrow \gamma\gamma$ channel with the ATLAS detector” (*Talk*).
- Deep Inelastic Scattering (DIS) 2025, Cape Town, South Africa. (24–28 March 2025), “Measurements of Higgs boson properties (mass, width, and Spin/CP) with the ATLAS detector” on behalf of the ATLAS collaboration (*Invited Talk*).

ATLAS Meetings and Workshops

- ATLAS Software & Computing Week #83, CERN. (9–13 February 2026), “FastCaloSim Group Progress Report” (*Invited Talk*).
- ATLAS UK Annual Meeting, Sussex. (7–9 January 2026), “Software & Computing Activities” on behalf of UK colleagues (*Invited Talk*).
- ATLAS Collaboration Week, CERN. (17–21 February 2025), “Beyond ISF: Muon Punch Through Simulation in AtlFast3” (*Lightning Talk*).
- ATLAS UK Annual Meeting, UCL. (6–8 January 2025), “Higgs to diphoton” analyses on behalf of UK colleagues (*Invited Talk*).

Teaching Experience

- Thermal Physics (PHYS09061). TA – Semester 2, 2025/2026
- Linear Algebra and Several Variable Calculus (PHYS08042). TA – Semester 1, 2025/2026
- Programming and Data Analysis: Python (PHYS08049). TA – Semester 1, 2025/2026
- Data Analysis and Machine Learning (PGPH11105). TA – Semester 2, 2024/2025
- Experimental Physics 2 (PHYS08058). TA – Semester 2, 2023/2024
- Physics 1B: The Stuff of the Universe (PHYS08017). TA – Semester 2, 2023/2024
- Relativity, Nuclear and Particle Physics (PHYS10096). TA – Semester 1, 2023/2024
- Physics 1A: Foundations (PHYS08016). TA – Semester 1, 2023/2024

Detector Operations

- Completed 376 hours of ATLAS detector operations shifts during a year-long attachment at CERN, including responsibilities in Run Control, Trigger, and Data Quality.